

CS5691: Pattern Recognition and Machine Learning

Programming Assignment 2

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Team - 25

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1 Dataset 1: Linearly Separable Data for 3 Classes

Dataset 1 contains 2-dimensional, linearly separable data for 3 classes. The following classifiers were used:

1.1 K-nearest Neighbors Classifier

The K-nearest neighbors (KNN) classifier was applied for three values of K : 1, 5, and 9. The distance metric used is Euclidean distance.

1.1.1 Results

The classification accuracies for different values of K are shown in Table 1.

K Value	Training Accuracy	Validation Accuracy	Test Accuracy
1	100.000%	98.333%	98.333%
5	98.571%	97.500%	98.333%
9	98.333%	97.500%	97.500%

Table 1: Classification Accuracies for KNN on Dataset 1

1.1.2 Confusion Matrix for Best Configuration

For $K = 1$, the confusion matrix on both the training and test datasets are shown in Figure 1. Both matrices are displayed side by side.

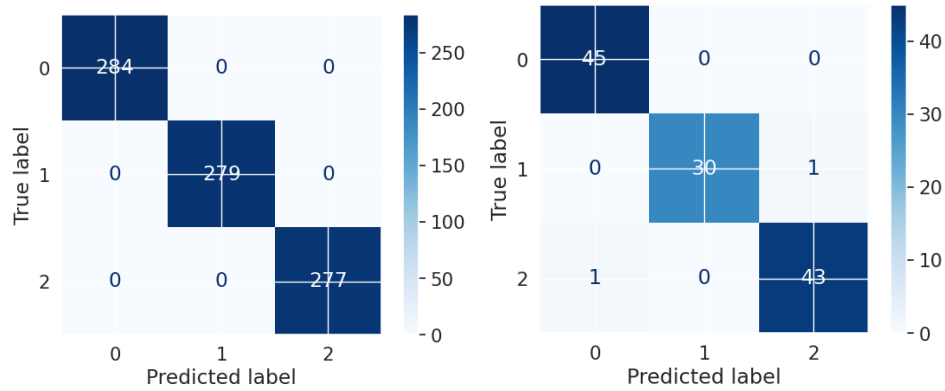


Figure 1: Confusion Matrices for KNN Classifier ($K = 1$) on Training (left) and Test (right) Data

1.1.3 Decision Region Plots

The decision region plots for $K = 1$, $K = 5$, and $K = 9$ are shown in Figures 2, 3, and 4, with the training data superimposed.

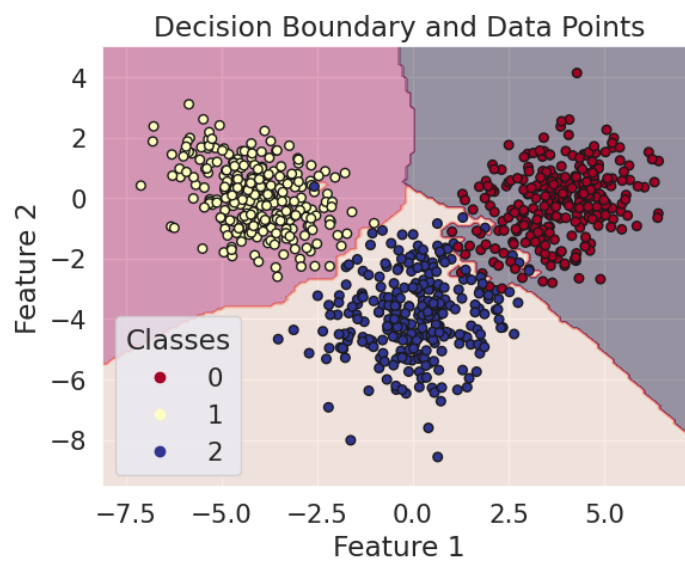


Figure 2: Decision Region Plot for KNN Classifier ($K = 1$)

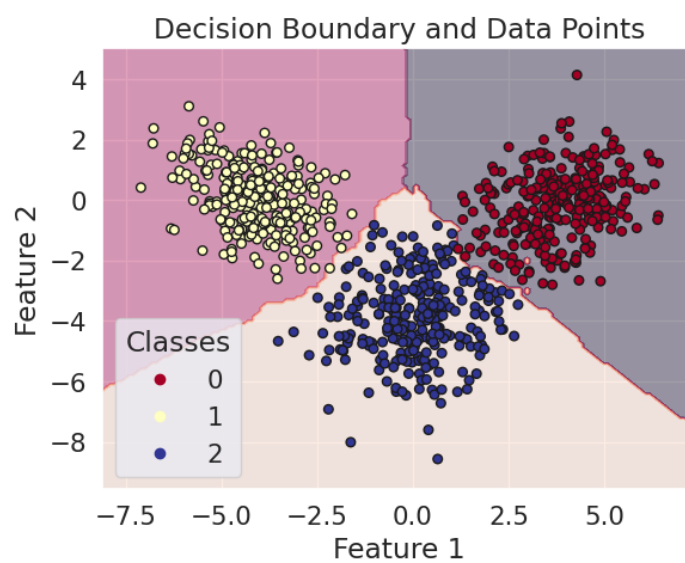


Figure 3: Decision Region Plot for KNN Classifier ($K = 5$)

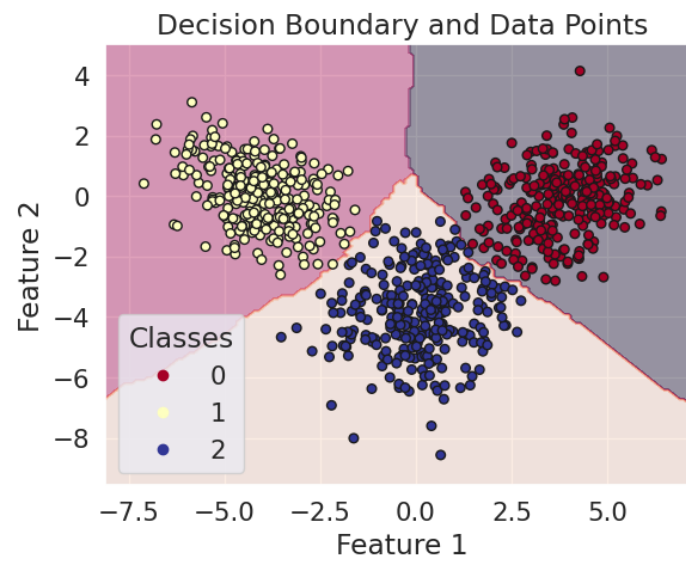


Figure 4: Decision Region Plot for KNN Classifier ($K = 9$)

1.2 Bayes Classifier with Gaussian Distribution when Covariance Matrices for all Classes are Same

1.2.1 Mathematical Formulation

In this scenario, each class C_l is assumed to have a Gaussian distribution with a class-specific mean vector μ_l but with a common covariance matrix Σ shared across all classes. The class-conditional density function for a feature vector $\mathbf{x}$ given class C_l is:

$$p(\mathbf{x}|C_l) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_l)^T \Sigma^{-1}(\mathbf{x} - \mu_l)\right)$$

where: - $\mathbf{x}$ is the feature vector, - μ_l is the mean vector of the Gaussian distribution for class C_l , - Σ is the common covariance matrix shared by all classes, - d is the dimensionality of the feature vector.

The prior probability $p(C_l)$ of each class C_l is estimated from the training data.

The posterior probability $p(C_l|\mathbf{x})$ is computed using Bayes' theorem:

$$p(C_l|\mathbf{x}) = \frac{p(\mathbf{x}|C_l)p(C_l)}{p(\mathbf{x})}$$

where $p(\mathbf{x})$ is the marginal likelihood of $\mathbf{x}$, given by:

$$p(\mathbf{x}) = \sum_{l=1}^l p(\mathbf{x}|C_l)p(C_l)$$

To simplify the classification task, we can use the discriminant function for each class. Since the covariance matrix Σ is the same for all classes, the discriminant function for class C_l is:

$$\delta_l(\mathbf{x}) = \log p(\mathbf{x}|C_l) + \log p(C_l)$$

Substituting $p(\mathbf{x}|C_l)$:

$$\delta_k(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \mu_l)^T \Sigma^{-1}(\mathbf{x} - \mu_l) - \frac{d}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma| + \log p(C_l)$$

The term $-\frac{d}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma|$ is constant for all classes and can be ignored when comparing the discriminant functions. Thus:

$$\delta_l(\mathbf{x}) \propto -\frac{1}{2}(\mathbf{x} - \mu_l)^T \Sigma^{-1}(\mathbf{x} - \mu_l) + \log p(C_l)$$

The classifier assigns the class label $\hat{y}$ to the feature vector $\mathbf{x}$ that maximizes the discriminant function:

$$\hat{y} = \arg \max_l \delta_l(\mathbf{x})$$

1.2.2 Results

The classification accuracies under this assumption are shown in Table 2.

Training Accuracy	Validation Accuracy	Test Accuracy
98.095%	98.333%	97.500%

Table 2: Classification Accuracy for Bayes Classifier (Same Covariance) on Dataset 1

1.2.3 Confusion Matrix for Best Configuration

The confusion matrices for both the training and test datasets are shown in Figure 5. Both matrices are displayed side by side.

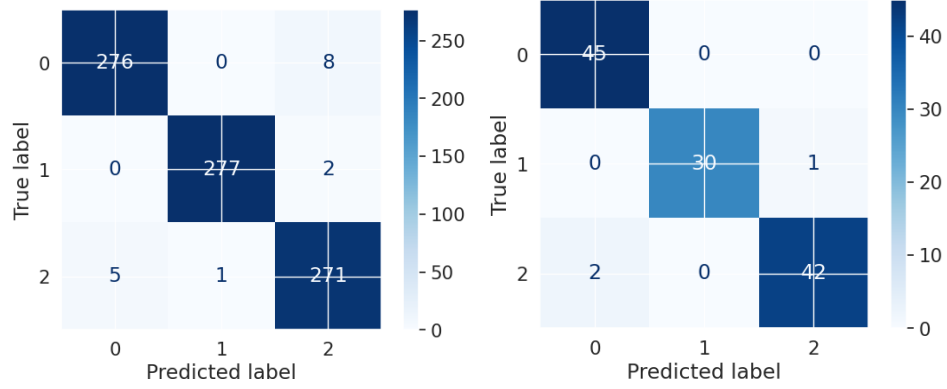


Figure 5: Confusion Matrices for Bayes Classifier (Same Covariance) on Training (left) and Test (right) Data

1.2.4 Decision Region Plot

The decision region plot for the Bayes classifier with the same covariance matrices is shown in Figure 6.

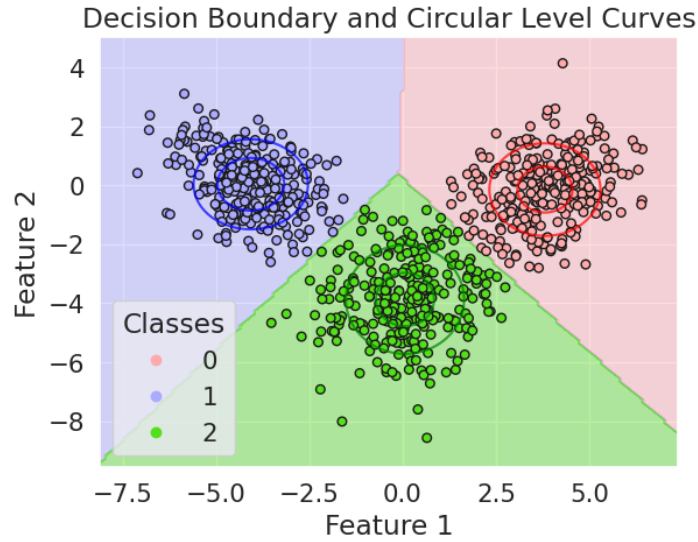


Figure 6: Decision Region Plot for Bayes Classifier (Same Covariance)

1.3 Bayes Classifier with Gaussian Distribution when Covariance Matrices for Each Class are Different

1.3.1 Mathematical Formulation

In this scenario, each class C_l is assumed to have its own Gaussian distribution with class-specific mean vector μ_l and covariance matrix Σ_l . The class-conditional density function for a feature vector $\mathbf{x}$ given class C_l is:

$$p(\mathbf{x}|C_l) = \frac{1}{(2\pi)^{d/2}|\Sigma_l|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_l)^T \Sigma_l^{-1}(\mathbf{x} - \mu_l)\right)$$

where: - $\mathbf{x}$ is the feature vector, - μ_l is the mean vector of the Gaussian distribution for class C_l , - Σ_l is the covariance matrix of the Gaussian distribution for class C_l , - d is the dimensionality of the feature vector.

The prior probability $p(C_l)$ of each class C_l is estimated from the training data.

The posterior probability $p(C_l|\mathbf{x})$ is computed using Bayes' theorem:

$$p(C_l|\mathbf{x}) = \frac{p(\mathbf{x}|C_l)p(C_l)}{p(\mathbf{x})}$$

where $p(\mathbf{x})$ is the marginal likelihood of $\mathbf{x}$, given by:

$$p(\mathbf{x}) = \sum_{l=1}^L p(\mathbf{x}|C_l)p(C_l)$$

The classifier assigns the class label $\hat{y}$ to the feature vector $\mathbf{x}$ that maximizes the posterior probability:

$$\hat{y} = \arg \max_l p(C_l|\mathbf{x})$$

1.3.2 Results

The classification accuracies under this assumption are shown in Table 3.

Training Accuracy	Validation Accuracy	Test Accuracy
98.214%	97.500%	98.333%

Table 3: Classification Accuracy for Bayes Classifier (Different Covariance) on Dataset 1

1.3.3 Confusion Matrix for Best Configuration

The confusion matrices for both the training and test datasets are shown in Figure 7. Both matrices are displayed side by side.

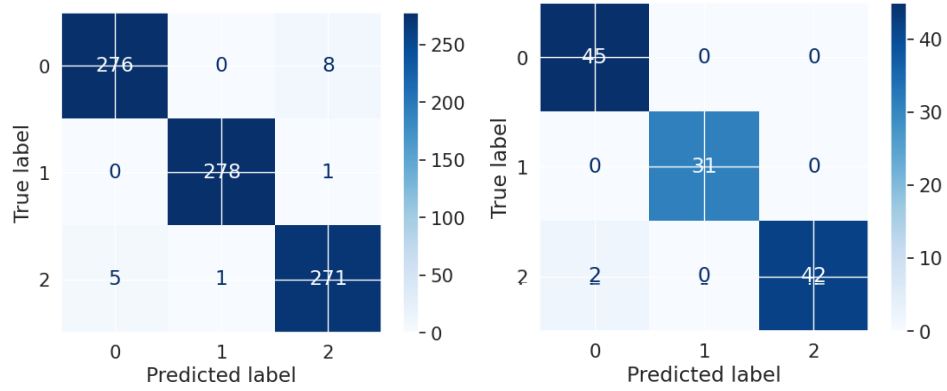


Figure 7: Confusion Matrices for Bayes Classifier (Different Covariance) on Training (left) and Test (right) Data

1.3.4 Level Curve and Decision Region Plot

The level curves and decision region plot for the Bayes classifier with different covariance matrices are shown in Figure 8.

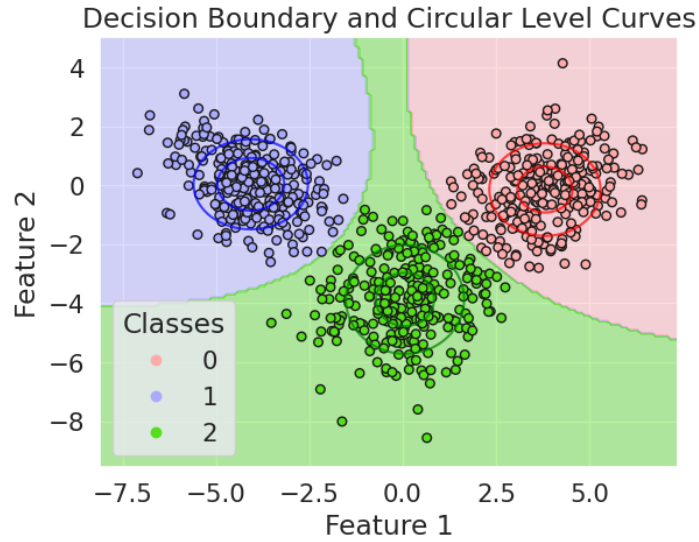


Figure 8: Level Curve and Decision Region Plot for Bayes Classifier (Different Covariance)

2 Dataset 2: Nonlinearly Separable Data for 2 Classes

Dataset 2 contains 2-dimensional, nonlinearly separable data for 2 classes. The following classifiers were applied:

2.1 K-nearest Neighbors Classifier

The K-nearest neighbors (KNN) classifier was applied for three values of K : 1, 5, and 9.

2.1.1 Results

The classification accuracies for different values of K are shown in Table 4.

K Value	Training Accuracy	Validation Accuracy	Test Accuracy
1	100.000%	100.000%	100.000%
5	99.821%	100.000%	100.000%
9	99.821%	100.000%	100.000%

Table 4: Classification Accuracies for KNN on Dataset 2

2.1.2 Confusion Matrix for Best Configuration

For $K = 1$, the confusion matrix on both the training and test datasets are shown in Figure 26. Both matrices are displayed side by side.

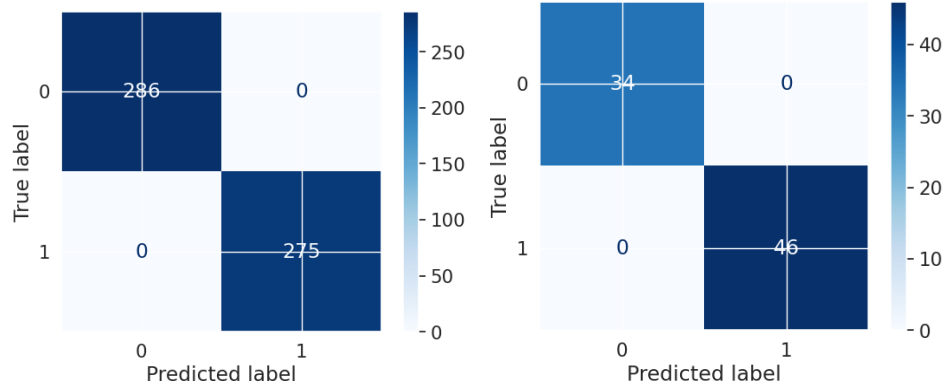


Figure 9: Confusion Matrices for KNN Classifier ($K = 1$) on Training (left) and Test (right) Data

2.1.3 Decision Region Plots

The decision region plots for $K = 1$, $K = 5$, and $K = 9$ are shown in Figures 10, 11, and 12, with the training data superimposed.

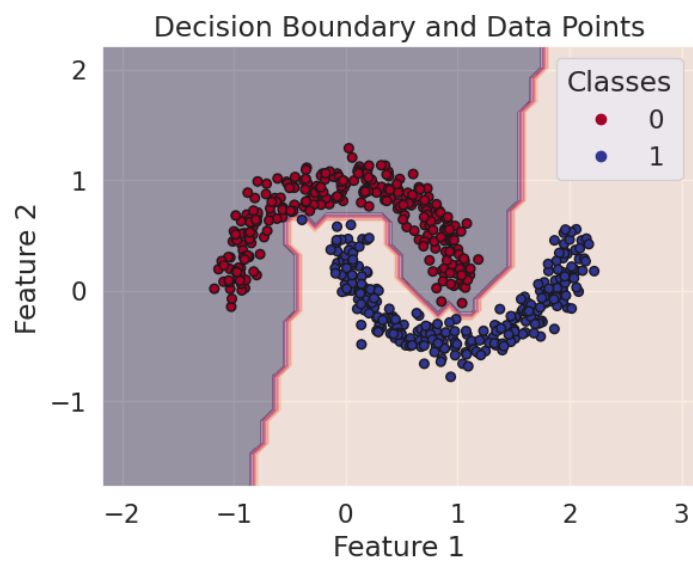


Figure 10: Decision Region Plot for KNN Classifier ($K = 1$)

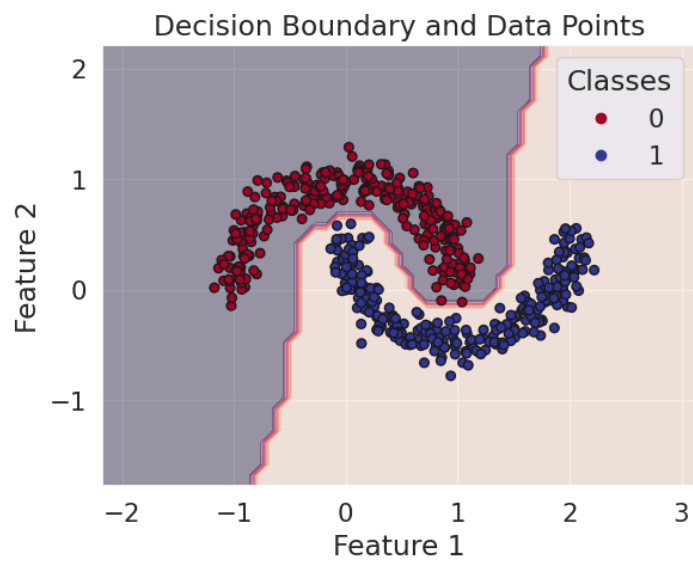


Figure 11: Decision Region Plot for KNN Classifier ($K = 5$)

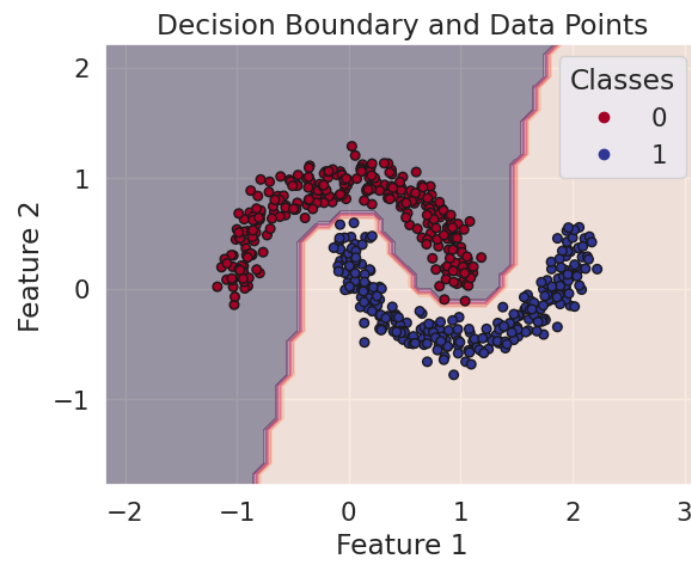


Figure 12: Decision Region Plot for KNN Classifier ($K = 9$)

2.2 K-nearest Representatives Classifier

The K-nearest representatives classifier was applied for three values of K : 1, 3, and 5, using 10 representatives per class.

2.2.1 Mathematical Formulation

This classifier is similar to KNN, but instead of using the nearest neighbors, the classifier assigns labels based on the K nearest representatives for each class. Each class is represented by 10 points (representatives).

2.2.2 Results

The classification accuracies for different values of K are shown in Table 5.

K Value	Training Accuracy	Validation Accuracy	Test Accuracy
1	99.822%	100.00%	100.00%
3	99.109%	98.742%	100.00%
5	97.326%	98.113%	97.500%

Table 5: Classification Accuracies for K-nearest Representatives on Dataset 2

2.2.3 Confusion Matrix for Best Configuration

For $K = 1$, the confusion matrix on both the training and test datasets are shown in Figure 26. Both matrices are displayed side by side.

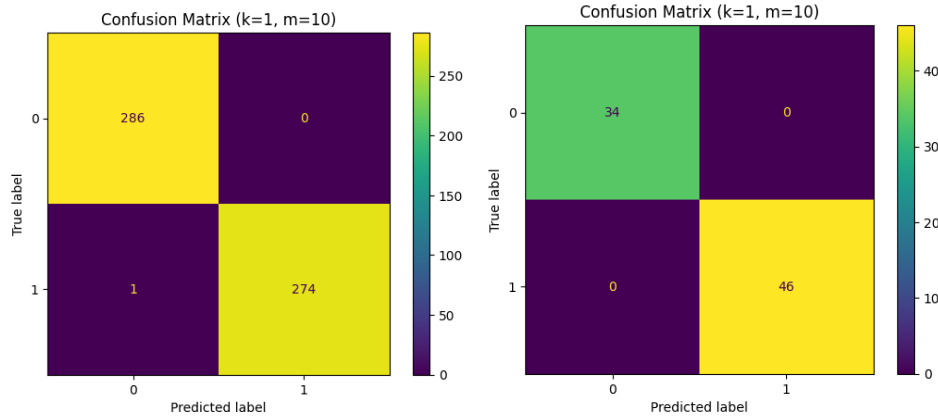


Figure 13: Confusion Matrices for K Nearest Representatives Classifier ($K = 1$) on Training (left) and Test (right) Data

2.2.4 Decision Region Plots

The decision region plots for $K = 1$, $K = 3$, and $K = 5$ are shown in Figures 14, 15, and 16.

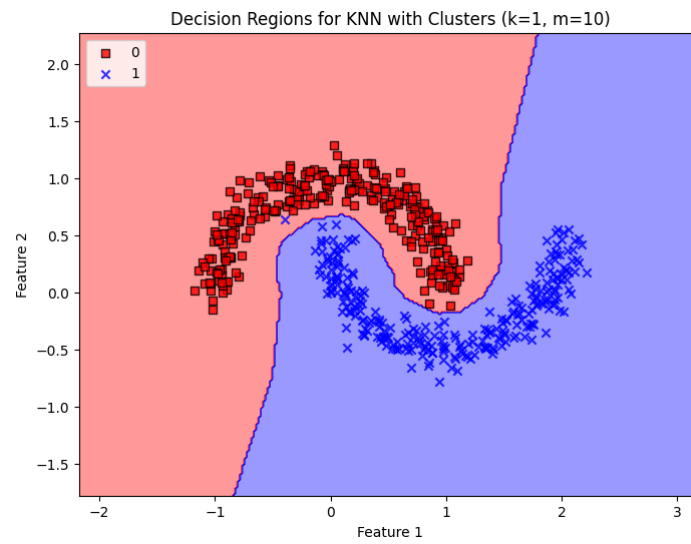


Figure 14: Decision Region Plot for K-nearest Representatives Classifier ($K = 1$)

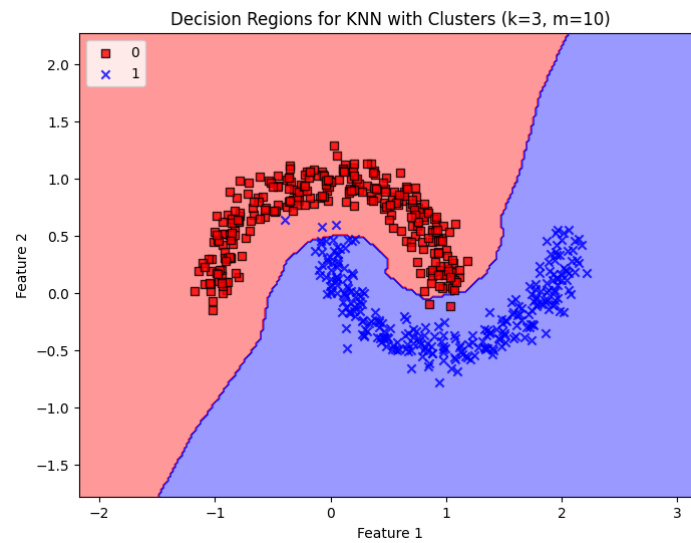


Figure 15: Decision Region Plot for K-nearest Representatives Classifier ($K = 3$)

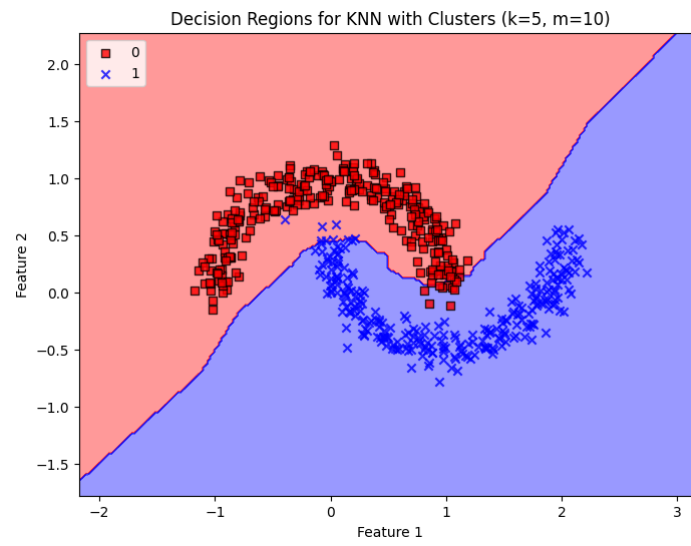


Figure 16: Decision Region Plot for K-nearest Representatives Classifier ($K = 5$)

2.3 Bayes Classifier with Gaussian Distribution when Covariance Matrices for all Classes are Same

2.3.1 Mathematical Formulation

In this scenario, each class C_l is assumed to have a Gaussian distribution with a class-specific mean vector μ_l but with a common covariance matrix Σ shared across all classes. The class-conditional density function for a feature vector $\mathbf{x}$ given class C_l is:

$$p(\mathbf{x}|C_l) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_l)^T \Sigma^{-1}(\mathbf{x} - \mu_l)\right)$$

where: - $\mathbf{x}$ is the feature vector, - μ_l is the mean vector of the Gaussian distribution for class C_l , - Σ is the common covariance matrix shared by all classes, - d is the dimensionality of the feature vector.

The prior probability $p(C_l)$ of each class C_l is estimated from the training data.

The posterior probability $p(C_l|\mathbf{x})$ is computed using Bayes' theorem:

$$p(C_l|\mathbf{x}) = \frac{p(\mathbf{x}|C_l)p(C_l)}{p(\mathbf{x})}$$

where $p(\mathbf{x})$ is the marginal likelihood of $\mathbf{x}$, given by:

$$p(\mathbf{x}) = \sum_{l=1}^l p(\mathbf{x}|C_l)p(C_l)$$

To simplify the classification task, we can use the discriminant function for each class. Since the covariance matrix Σ is the same for all classes, the discriminant function for class C_l is:

$$\delta_l(\mathbf{x}) = \log p(\mathbf{x}|C_l) + \log p(C_l)$$

Substituting $p(\mathbf{x}|C_l)$:

$$\delta_k(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \mu_l)^T \Sigma^{-1}(\mathbf{x} - \mu_l) - \frac{d}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma| + \log p(C_l)$$

The term $-\frac{d}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma|$ is constant for all classes and can be ignored when comparing the discriminant functions. Thus:

$$\delta_l(\mathbf{x}) \propto -\frac{1}{2}(\mathbf{x} - \mu_l)^T \Sigma^{-1}(\mathbf{x} - \mu_l) + \log p(C_l)$$

The classifier assigns the class label $\hat{y}$ to the feature vector $\mathbf{x}$ that maximizes the discriminant function:

$$\hat{y} = \arg \max_l \delta_l(\mathbf{x})$$

2.3.2 Results

The classification accuracy under this assumption are shown in Table 6.

Training Accuracy	Validation Accuracy	Test Accuracy
87.165%	90.566%	88.750%

Table 6: Classification Accuracy for Bayes Classifier (Same Covariance) on Dataset 2

2.3.3 Confusion Matrix for Best Configuration

The confusion matrices for both the training and test datasets are shown in Figure 17.

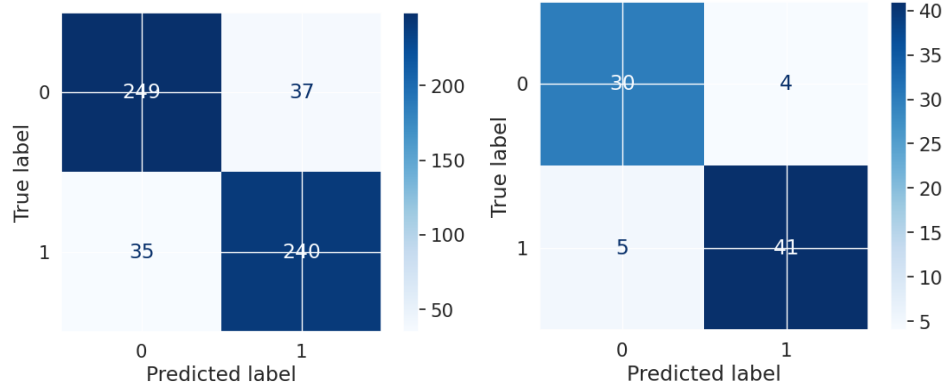


Figure 17: Confusion Matrices for Bayes Classifier (Same Covariance) on Training (left) and Test (right) Data

2.3.4 Decision Region Plot

The decision region plot for the Bayes classifier with the same covariance matrices is shown in Figure 18.

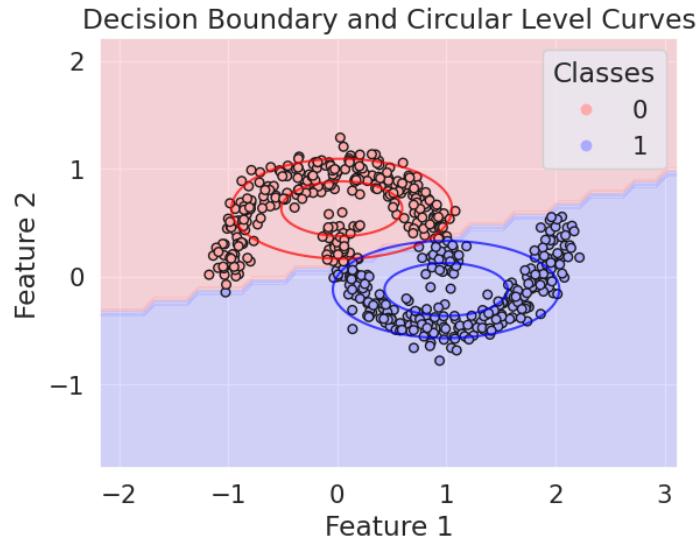


Figure 18: Decision Region Plot for Bayes Classifier (Same Covariance)

2.3.5 Mathematical Formulation

In this scenario, each class C_l is assumed to have its own Gaussian distribution with class-specific mean vector μ_l and covariance matrix Σ_l . The class-conditional density function for a feature vector $\mathbf{x}$ given class C_l is:

$$p(\mathbf{x}|C_l) = \frac{1}{(2\pi)^{d/2}|\Sigma_l|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_l)^T \Sigma_l^{-1}(\mathbf{x} - \mu_l)\right)$$

where: - $\mathbf{x}$ is the feature vector, - μ_l is the mean vector of the Gaussian distribution for class C_l , - Σ_l is the covariance matrix of the Gaussian distribution for class C_l , - d is the dimensionality of the feature vector.

The prior probability $p(C_l)$ of each class C_l is estimated from the training data.

The posterior probability $p(C_l|\mathbf{x})$ is computed using Bayes' theorem:

$$p(C_l|\mathbf{x}) = \frac{p(\mathbf{x}|C_l)p(C_l)}{p(\mathbf{x})}$$

where $p(\mathbf{x})$ is the marginal likelihood of $\mathbf{x}$, given by:

$$p(\mathbf{x}) = \sum_{l=1}^L p(\mathbf{x}|C_l)p(C_l)$$

The classifier assigns the class label $\hat{y}$ to the feature vector $\mathbf{x}$ that maximizes the posterior probability:

$$\hat{y} = \arg \max_l p(C_l|\mathbf{x})$$

2.3.6 Results

The classification accuracy under this assumption are shown in Table 7.

Training Accuracy	Validation Accuracy	Test Accuracy
86.987%	90.566%	88.750%

Table 7: Classification Accuracy for Bayes Classifier (Different Covariance) on Dataset 2

2.3.7 Confusion Matrix for Best Configuration

The confusion matrices for both the training and test datasets are shown in Figure 19.

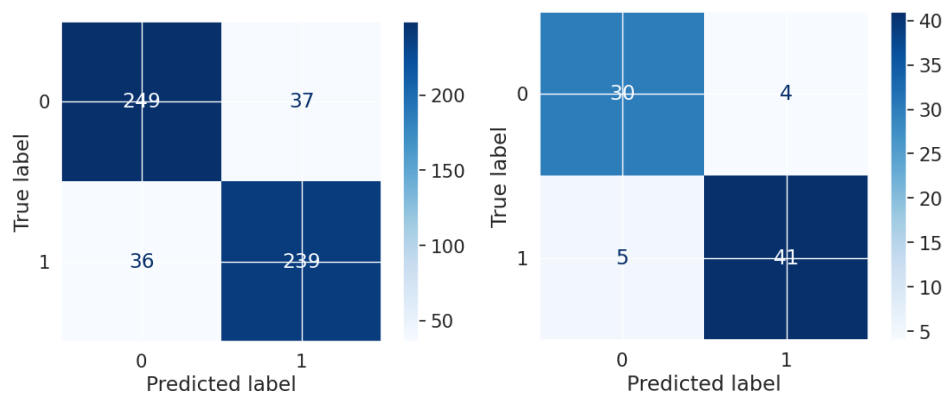


Figure 19: Confusion Matrices for Bayes Classifier (Different Covariance) on Training (left) and Test (right) Data

2.3.8 Decision Region Plot

The decision region plot for the Bayes classifier with different covariance matrices is shown in Figure 20.

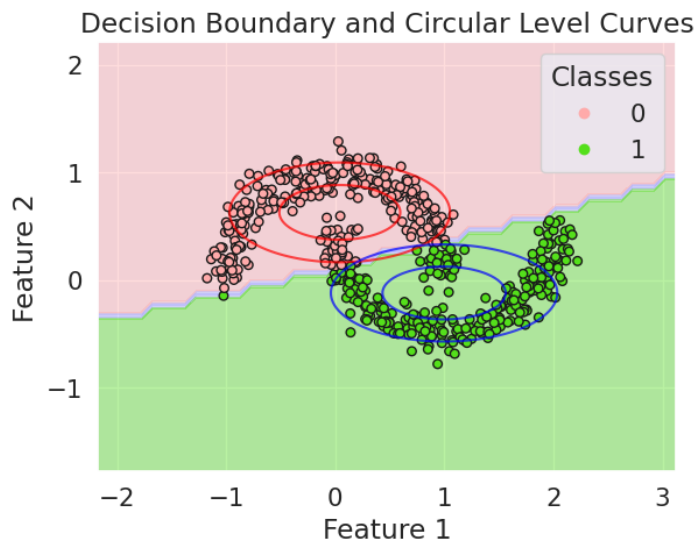


Figure 20: Decision Region Plot for Bayes Classifier (Different Covariance)

2.4 Naive-Bayes Classifier with Gaussian Distribution when Covariance Matrices for all Classes are Same

2.4.1 Mathematical Formulation

In this case, the Naive-Bayes classifier assumes that features are conditionally independent given the class, and each class is modeled with a Gaussian distribution where all classes share the same covariance matrix Σ .

Let C_l be the class label, $\mathbf{x} = (x_1, x_2, \dots, x_d)^T$ be the feature vector, and $\mu_l = (\mu_{l1}, \mu_{l2}, \dots, \mu_{ld})^T$ be the mean vector for class C_l . The class-conditional density function for the feature vector $\mathbf{x}$ given class C_l is:

$$p(\mathbf{x}|C_l) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_l)^T \Sigma^{-1}(\mathbf{x} - \mu_l)\right)$$

where: - Σ is the shared covariance matrix for all classes, - μ_l is the mean vector of the Gaussian distribution for class C_l , - d is the dimensionality of the feature vector.

Given the Naive-Bayes assumption that features are conditionally independent given the class, the joint probability of observing feature vector $\mathbf{x}$ is:

$$p(\mathbf{x}|C_l) = \prod_{j=1}^d \frac{1}{\sqrt{2\pi\sigma_j^2}} \exp\left(-\frac{(x_j - \mu_{lj})^2}{2\sigma_j^2}\right)$$

where σ_j^2 is the variance of feature x_j and is the same for all classes.

The prior probability $p(C_l)$ of each class C_l is estimated from the training data.

The posterior probability $p(C_l|\mathbf{x})$ is computed using Bayes' theorem:

$$p(C_l|\mathbf{x}) = \frac{p(\mathbf{x}|C_l)p(C_l)}{p(\mathbf{x})}$$

where $p(\mathbf{x})$ is the marginal likelihood of $\mathbf{x}$, given by:

$$p(\mathbf{x}) = \sum_{l=1}^L p(\mathbf{x}|C_l)p(C_l)$$

The classifier assigns the class label $\hat{y}$ to the feature vector $\mathbf{x}$ that maximizes the posterior probability:

$$\hat{y} = \arg \max_l p(C_l|\mathbf{x})$$

2.4.2 Results

The classification accuracy under this assumption are shown in Table 8.

Training Accuracy	Validation Accuracy	Test Accuracy
86.987%	90.566%	88.750%

Table 8: Classification Accuracy for Naive-Bayes Classifier (Same Covariance) on Dataset 2

2.4.3 Confusion Matrix for Best Configuration

The confusion matrices for both the training and test datasets are shown in Figure 21.

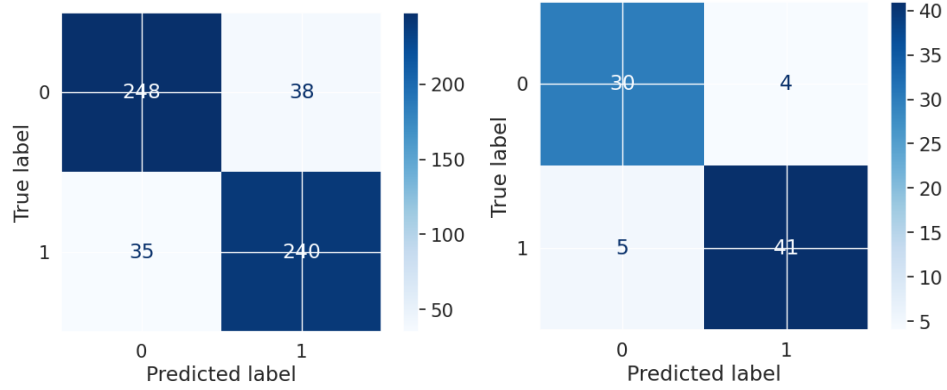


Figure 21: Confusion Matrices for Naive-Bayes Classifier (Same Covariance) on Training (left) and Test (right) Data

2.4.4 Decision Region Plot

The decision region plot for the Naive-Bayes classifier with the same covariance matrices is shown in Figure 22.

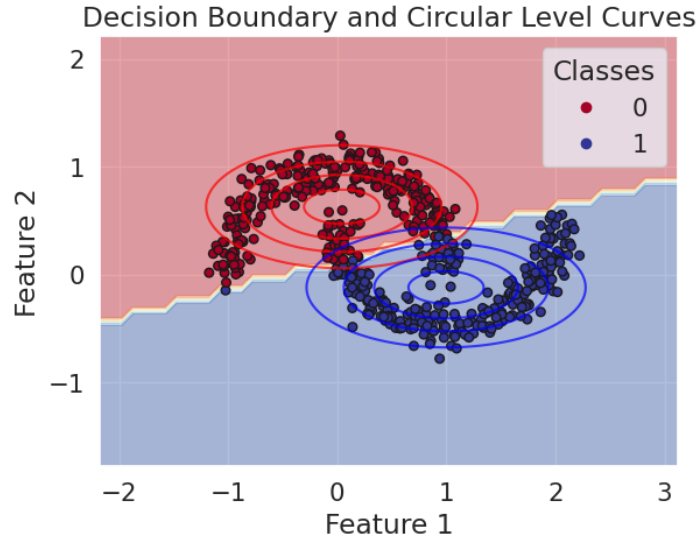


Figure 22: Decision Region Plot for Naive-Bayes Classifier (Same Covariance)

2.5 Naive-Bayes Classifier with Gaussian Distribution when Covariance Matrices for Each Class are Different

2.5.1 Mathematical Formulation

In the Naive-Bayes classifier, the assumption is that the features are conditionally independent given the class label. For the case where each class has its own Gaussian distribution with distinct covariance matrices, the class-conditional density function $p(\mathbf{x}|C_l)$ is given by:

$$p(\mathbf{x}|C_l) = \prod_{j=1}^d \frac{1}{\sqrt{2\pi\sigma_{lj}^2}} \exp\left(-\frac{(x_j - \mu_{lj})^2}{2\sigma_{lj}^2}\right)$$

where: - $\mathbf{x} = (x_1, x_2, \dots, x_d)^T$ is the feature vector, - μ_{lj} is the mean of the j -th feature for class C_l , - σ_{lj}^2 is the variance of the j -th feature for class C_l , - d is the number of features.

The prior probability $p(C_l)$ of each class C_l is estimated from the training data.

Using Bayes' theorem, the posterior probability $p(C_l|\mathbf{x})$ is:

$$p(C_l|\mathbf{x}) = \frac{p(\mathbf{x}|C_l)p(C_l)}{p(\mathbf{x})}$$

where the marginal likelihood $p(\mathbf{x})$ is:

$$p(\mathbf{x}) = \sum_{l=1}^L p(\mathbf{x}|C_l)p(C_l)$$

The Naive-Bayes classifier assigns the class label $\hat{y}$ to the feature vector $\mathbf{x}$ that maximizes the posterior probability:

$$\hat{y} = \arg \max_l p(C_l|\mathbf{x})$$

Given the independence assumption, the posterior probability simplifies to:

$$p(C_l|\mathbf{x}) \propto p(\mathbf{x}|C_l)p(C_l)$$

Thus, the classifier assigns $\hat{y}$ based on:

$$\hat{y} = \arg \max_l \left[\prod_{j=1}^d \frac{1}{\sqrt{2\pi\sigma_{lj}^2}} \exp\left(-\frac{(x_j - \mu_{lj})^2}{2\sigma_{lj}^2}\right) \cdot p(C_l) \right]$$

2.5.2 Results

The classification accuracy under this assumption are shown in Table 9.

Training Accuracy	Validation Accuracy	Test Accuracy
87.165%	90.566%	88.750%

Table 9: Classification Accuracies for Naive-Bayes Classifier (Different Covariance) on Dataset 2

2.5.3 Confusion Matrix for Best Configuration

The confusion matrices for both the training and test datasets are shown in Figure 23.

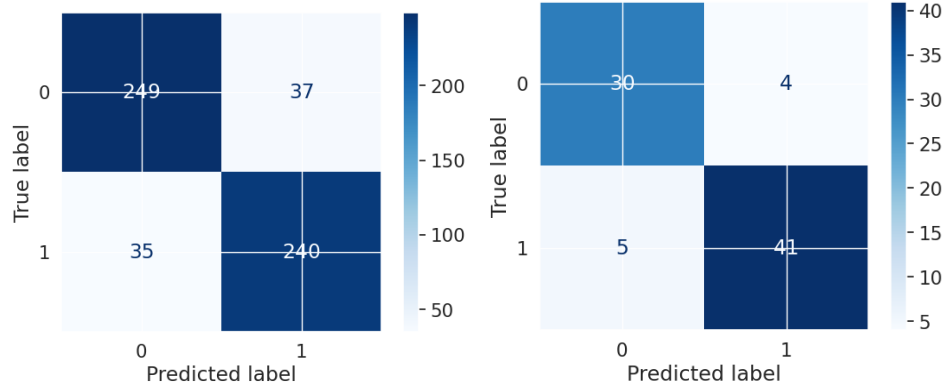


Figure 23: Confusion Matrices for Naive-Bayes Classifier (Different Covariance) on Training (left) and Test (right) Data

2.5.4 Decision Region Plot

The decision region plot for the Naive-Bayes classifier with different covariance matrices is shown in Figure 24.

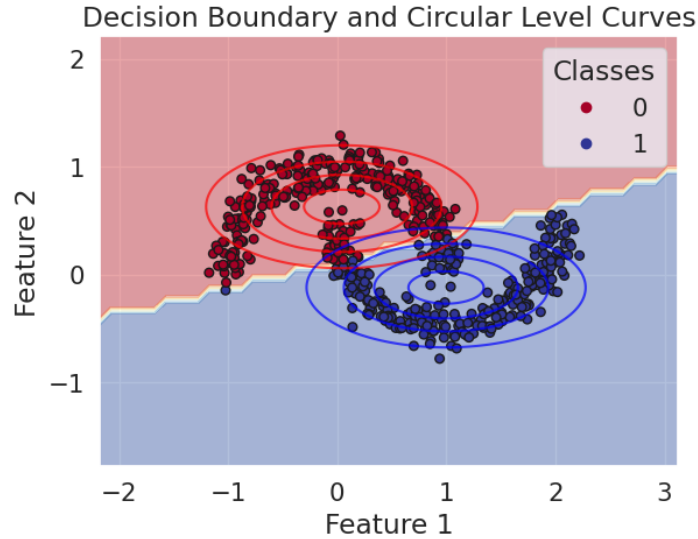


Figure 24: Decision Region Plot for Naive-Bayes Classifier (Different Covariance)

3 Dataset 3: Image Data for 5 Classes

Dataset 3 contains image data with a feature vector dimension of 35 for 5 classes. The following classifiers were applied:

3.1 K-nearest Neighbors Classifier

The K-nearest neighbors (KNN) classifier was applied for three values of K : 1, 9, and 15.

3.1.1 Results

The classification accuracy for different values of K are shown in Table 10.

K Value	Training Accuracy	Validation Accuracy	Test Accuracy
1	100.000%	61.204%	55.927%
9	72.606%	67.224%	62.103%
15	69.842%	67.558%	61.102%

Table 10: Classification Accuracie for KNN on Dataset 3

3.1.2 Confusion Matrix for Best Configuration

For $K = 15$, the confusion matrices on both the training and test datasets are shown in Figure 25. Both matrices are displayed side by side.

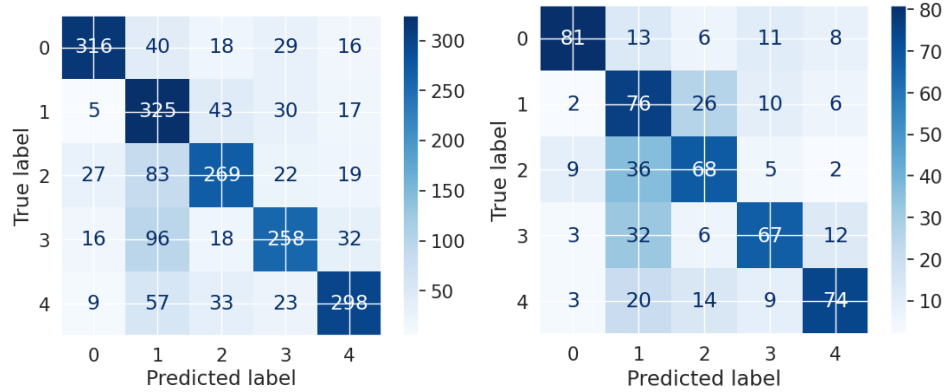


Figure 25: Confusion Matrices for KNN Classifier ($K = 15$) on Training (left) and Test (right) Data

3.2 K-nearest Representatives Classifier

The K-nearest representatives classifier was applied for three values of K : 1, 5, and 9, using 10 representatives per class.

3.2.1 Mathematical Formulation

This classifier is similar to KNN, but instead of using the nearest neighbors, the classifier assigns labels based on the K nearest representatives for each class. Each class is represented by 10 points (representatives).

3.2.2 Results

The classification accuracies for different values of K are shown in Table 11.

K Value	Training Accuracy	Validation Accuracy	Test Accuracy
1	64.883%	64.883%	59.933%
5	62.207%	62.207%	57.595%
9	61.204%	61.204%	56.594%

Table 11: Classification Accuracies for K-nearest Representatives on Dataset 3

3.2.3 Confusion Matrix for Best Configuration

For $K = 1$, the confusion matrix on both the training and test datasets are shown in Figure 26. Both matrices are displayed side by side.

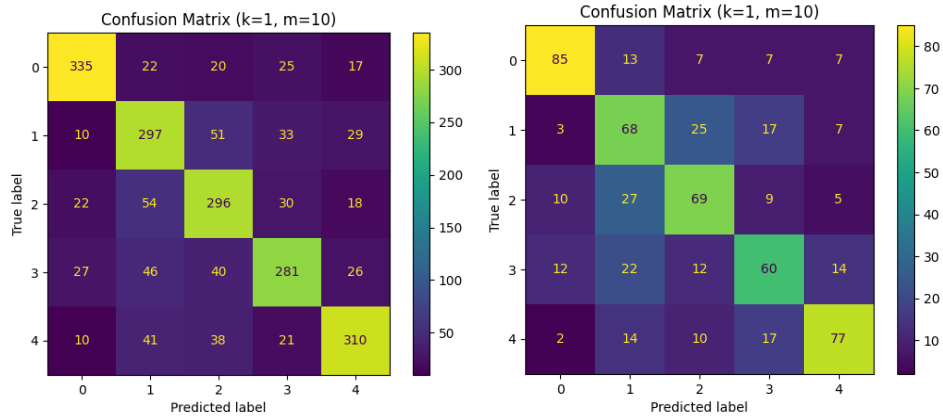


Figure 26: Confusion Matrices for K Nearest Representatives Classifier ($K = 1$) on Training (left) and Test (right) Data

3.3 Bayes Classifier with Gaussian Distribution

The Bayes classifier with a Gaussian distribution was applied. Results are presented when the covariance matrices are assumed to be different for each class.

3.3.1 Mathematical Formulation

In this scenario, each class C_l is assumed to have its own Gaussian distribution with class-specific mean vector μ_l and covariance matrix Σ_l . The class-conditional density function for a feature vector $\mathbf{x}$ given class C_l is:

$$p(\mathbf{x}|C_l) = \frac{1}{(2\pi)^{d/2}|\Sigma_l|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_l)^T \Sigma_l^{-1}(\mathbf{x} - \mu_l)\right)$$

where: - $\mathbf{x}$ is the feature vector, - μ_l is the mean vector of the Gaussian distribution for class C_l , - Σ_l is the covariance matrix of the Gaussian distribution for class C_l , - d is the dimensionality of the feature vector.

The prior probability $p(C_l)$ of each class C_l is estimated from the training data.

The posterior probability $p(C_l|\mathbf{x})$ is computed using Bayes' theorem:

$$p(C_l|\mathbf{x}) = \frac{p(\mathbf{x}|C_l)p(C_l)}{p(\mathbf{x})}$$

where $p(\mathbf{x})$ is the marginal likelihood of $\mathbf{x}$, given by:

$$p(\mathbf{x}) = \sum_{l=1}^L p(\mathbf{x}|C_l)p(C_l)$$

The classifier assigns the class label $\hat{y}$ to the feature vector $\mathbf{x}$ that maximizes the posterior probability:

$$\hat{y} = \arg \max_l p(C_l|\mathbf{x})$$

3.3.2 Results

The classification accuracies under this assumption are shown in Table 12.

Training Accuracy	Validation Accuracy	Test Accuracy
73.225%	56.187%	57.429%

Table 12: Classification Accuracies for Bayes Classifier (Different Covariance) on Dataset 3

3.3.3 Confusion Matrix for Best Configuration

The confusion matrices for both the training and test datasets are shown in Figure 27.

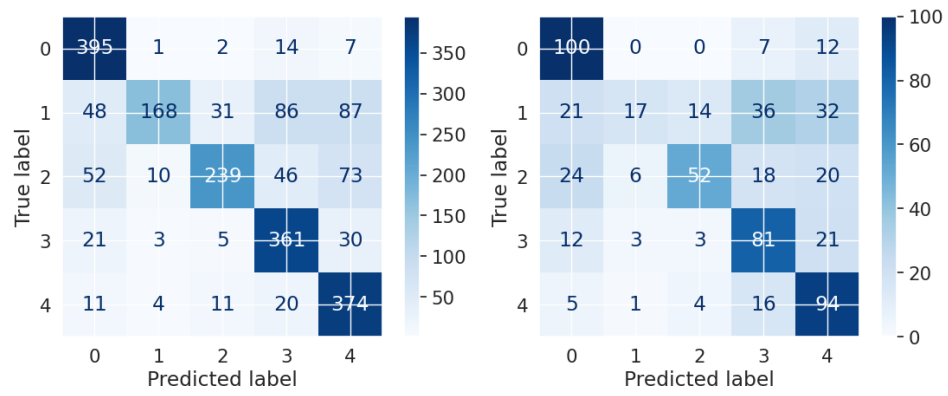


Figure 27: Confusion Matrices for Bayes Classifier (Different Covariance) on Training (left) and Test (right) Data

3.4 Naive-Bayes Classifier with Gaussian Distribution

3.4.1 Mathematical Formulation

In the Naive-Bayes classifier, the assumption is that the features are conditionally independent given the class label. For the case where each class has its own Gaussian distribution with distinct covariance matrices, the class-conditional density function $p(\mathbf{x}|C_l)$ is given by:

$$p(\mathbf{x}|C_l) = \prod_{j=1}^d \frac{1}{\sqrt{2\pi\sigma_{lj}^2}} \exp\left(-\frac{(x_j - \mu_{lj})^2}{2\sigma_{lj}^2}\right)$$

where: - $\mathbf{x} = (x_1, x_2, \dots, x_d)^T$ is the feature vector, - μ_{lj} is the mean of the j -th feature for class C_l , - σ_{lj}^2 is the variance of the j -th feature for class C_l , - d is the number of features.

The prior probability $p(C_l)$ of each class C_l is estimated from the training data.

Using Bayes' theorem, the posterior probability $p(C_l|\mathbf{x})$ is:

$$p(C_l|\mathbf{x}) = \frac{p(\mathbf{x}|C_l)p(C_l)}{p(\mathbf{x})}$$

where the marginal likelihood $p(\mathbf{x})$ is:

$$p(\mathbf{x}) = \sum_{l=1}^L p(\mathbf{x}|C_l)p(C_l)$$

The Naive-Bayes classifier assigns the class label $\hat{y}$ to the feature vector $\mathbf{x}$ that maximizes the posterior probability:

$$\hat{y} = \arg \max_l p(C_l|\mathbf{x})$$

Given the independence assumption, the posterior probability simplifies to:

$$p(C_l|\mathbf{x}) \propto p(\mathbf{x}|C_l)p(C_l)$$

Thus, the classifier assigns $\hat{y}$ based on:

$$\hat{y} = \arg \max_l \left[\prod_{j=1}^d \frac{1}{\sqrt{2\pi\sigma_{lj}^2}} \exp\left(-\frac{(x_j - \mu_{lj})^2}{2\sigma_{lj}^2}\right) \cdot p(C_l) \right]$$

3.4.2 Results

The classification accuracies under this assumption are shown in Table 13.

Training Accuracy	Validation Accuracy	Test Accuracy
58.837%	58.862%	56.093%

Table 13: Classification Accuracies for Naive Bayes Classifier (Different Covariance) on Dataset 3

3.4.3 Confusion Matrix for Best Configuration

The confusion matrices for both the training and test datasets are shown in Figure 28.

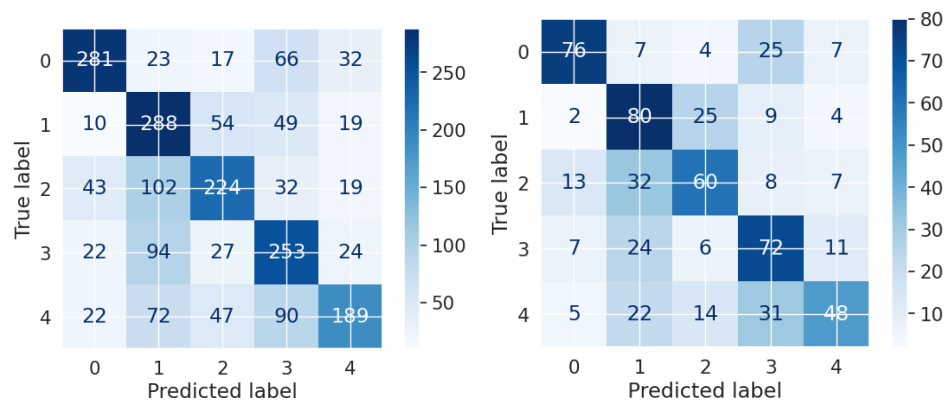


Figure 28: Confusion Matrices for Naive-Bayes Classifier (Different Covariance) on Training (left) and Test (right) Data

4 Conclusion

The classifiers were evaluated across three datasets, and their performance metrics and visualizations were documented. Each classifier demonstrated unique strengths and weaknesses depending on the dataset and the parameters used.